



Edexcel IGCSE Physics



Your notes

Current, Potential Difference & Resistance

Contents

- * Charge & Current
- * Voltage & Energy
- * Resistance



Your notes

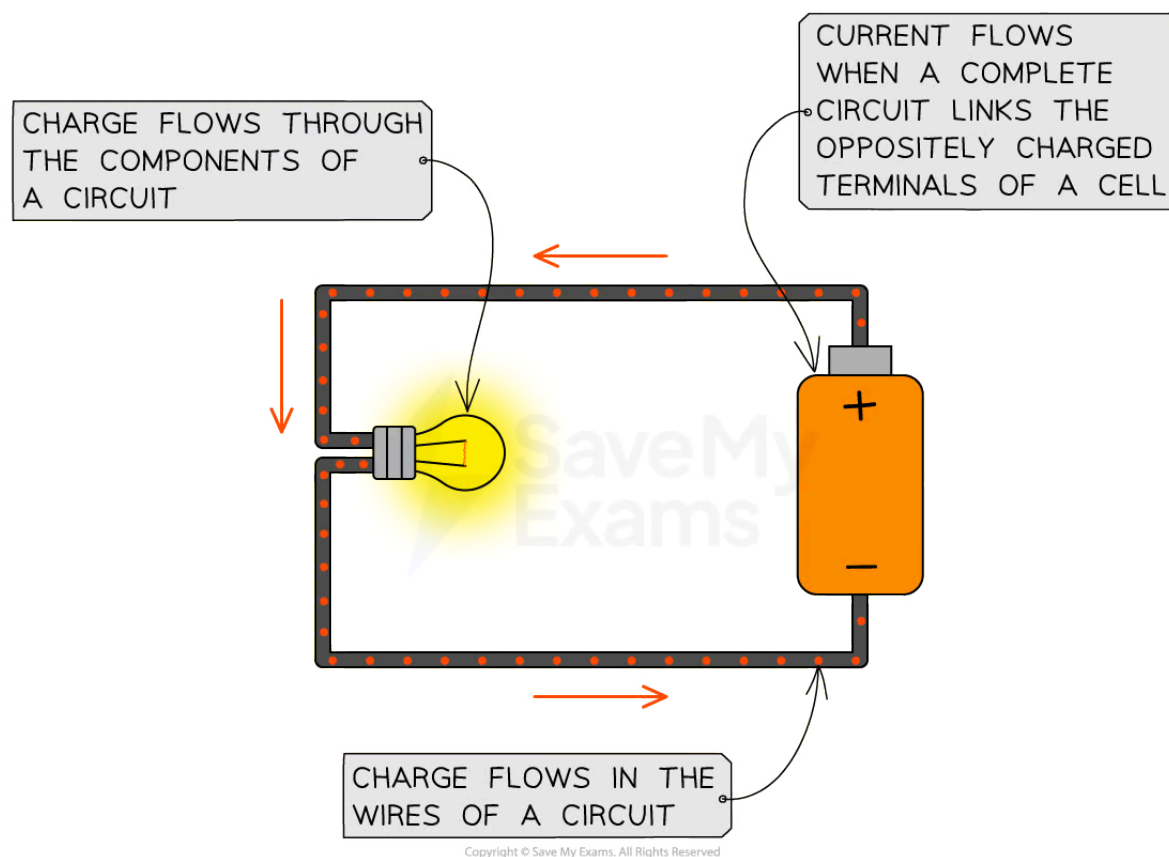
Charge & Current

Current

- Electric current is defined as

The rate of flow of electric charge

- Current is measured in units of **amperes** or **amps (A)**
 - 1 amp is equivalent to a charge of 1 coulomb flowing in 1 second, or $1\text{ A} = 1\text{ C/s}$
- This means the size of an electric current is the amount of charge passing through a component each second
- Current flows
 - when a **circuit** is formed e.g. when a wire connects the two oppositely charged terminals of a cell
 - from the **positive** terminal to the **negative** terminal of a cell

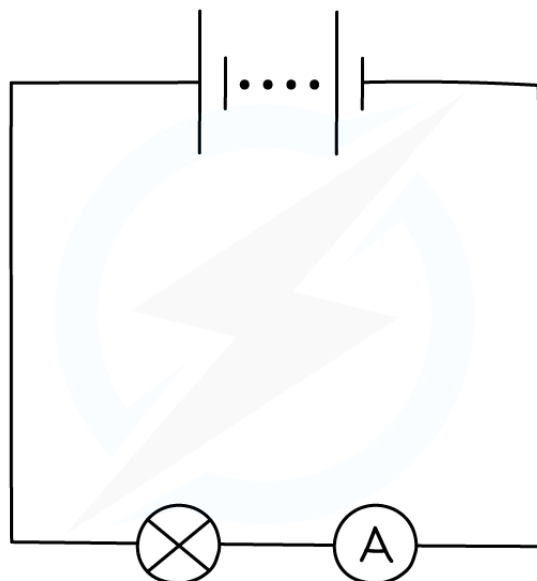


Charge flows from the positive terminal to the negative terminal

Measuring current

- Current can be measured using an **ammeter**

- Ammeters must be connected in **series** with the component being measured



Copyright © Save My Exams. All Rights Reserved

An ammeter can be used to measure the current around a circuit



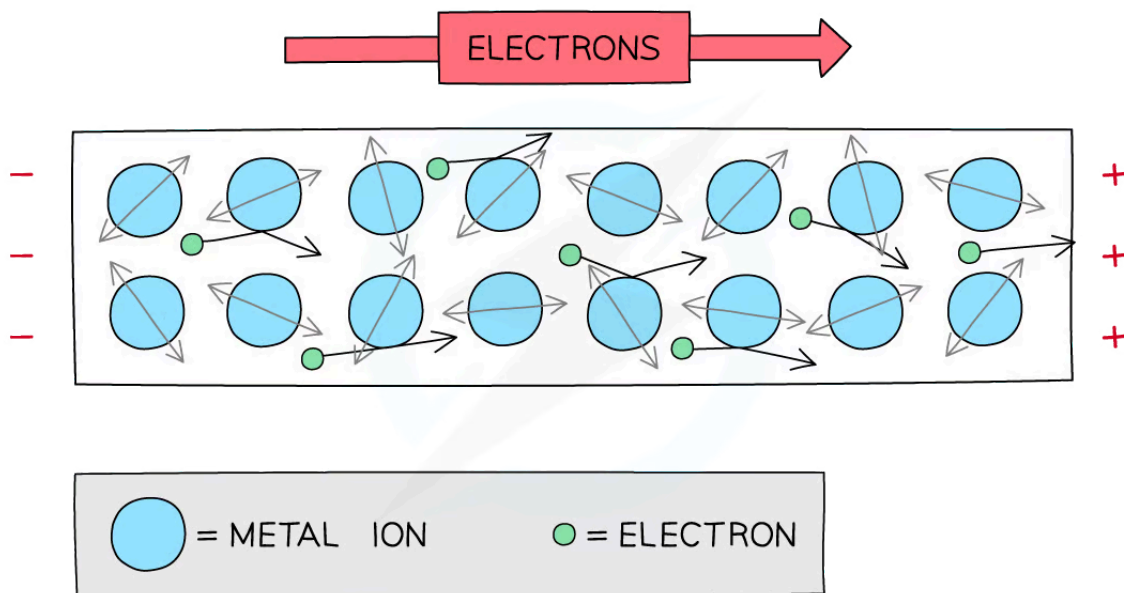
Your notes

Charge

- The wires in an electric circuit are made of **metal** because it is a good **conductor** of electric **current**
- In the wires, the current is a flow of **negatively charged electrons**



Your notes



Copyright © Save My Exams. All Rights Reserved



In metal wires, the current is a flow of negatively charged electrons. When a voltage is applied, electrons flow through the lattice of metal ions

Examiner Tip

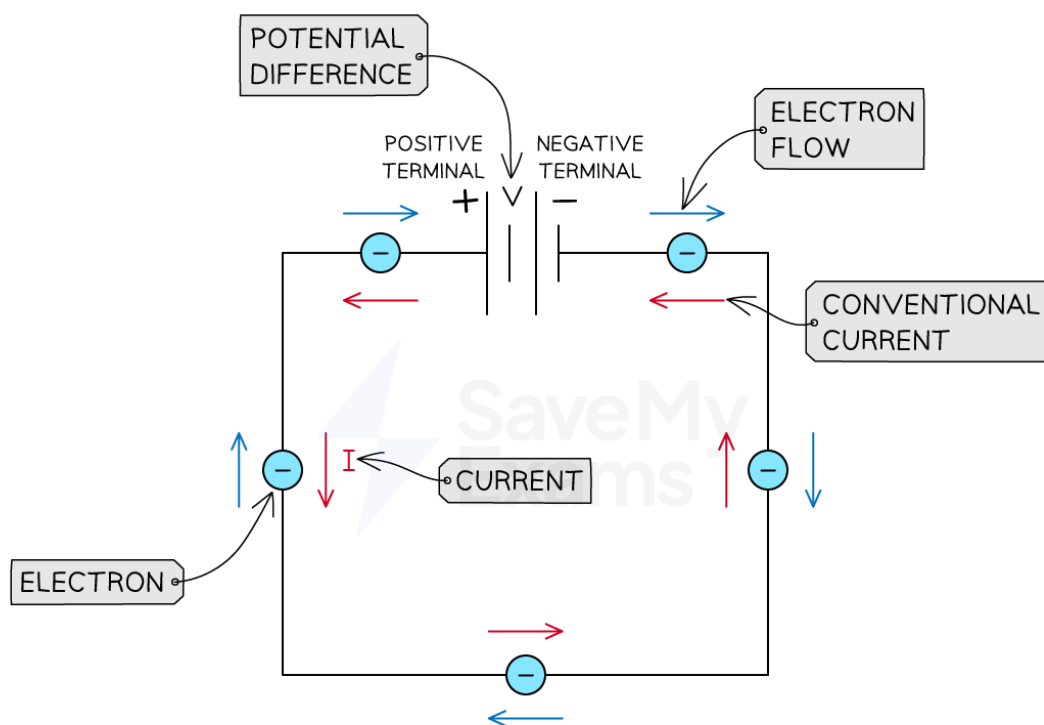
You should always consider current to be the flow of **positive** charge i.e. from the positive terminal to the negative terminal of a cell. This is known as **conventional current**.

This is in the opposite direction to **electron flow**, which is the flow of negatively charged electrons from the negative terminal to the positive terminal of a cell.

This is the convention we use because scientists defined conventional current before they discovered the electron



Your notes



Copyright © Save My Exams. All Rights Reserved



Your notes

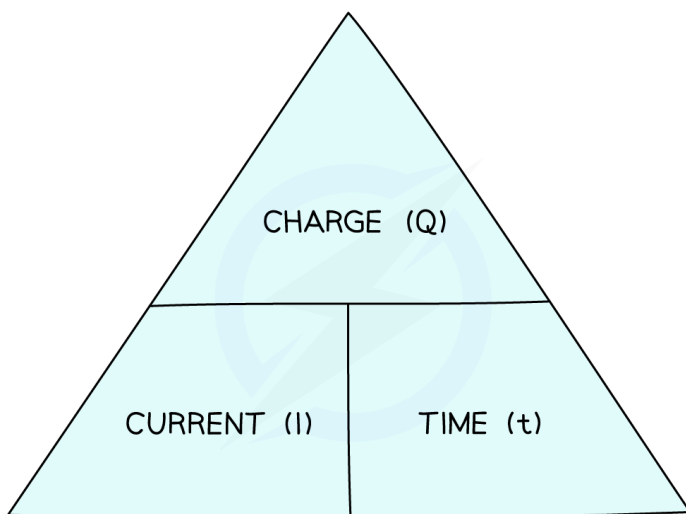
Calculating electric charge

- Current, charge and time are related by the equation:
charge = current $\times$ time

$$Q = I \times t$$

- Where:
 - Q = charge, measured in coulombs (C)
 - I = current, measured in amps (A)
 - t = time, measured in seconds (s)
- The current, charge and time equation can be rearranged with the help of the following formula triangle:

Current charge time formula triangle

Copyright © Save My Exams. All Rights Reserved

Formula triangle for the charge, current and time equation



Your notes

 Worked example

When will 8 A of current pass through an electrical circuit?

- A. When 8 J of energy is used by 1 C of charge
- B. When a charge of 4 C passes in 0.5 s
- C. When a charge of 8 C passes in 0.1 s
- D. When a charge of 1 C passes in 8 s

ANSWER: B

- The equation relating current, charge and time is:

$$Q = I \times t$$

- Rearrange to make current I the subject of the equation:

$$I = \frac{Q}{t}$$

- Consider option **B**, where $Q = 4 \text{ C}$ and $t = 0.5 \text{ s}$:

$$I = \frac{4}{0.5} = 8 \text{ A}$$

- Therefore, the correct answer is **B**

A is incorrect as this is the definition of a voltage of 8 V between two points and does not describe current

C is incorrect as $I = \frac{8}{0.1} = 80 \text{ A}$

D is incorrect as $I = \frac{1}{8} = 0.125 \text{ A}$

 Examiner Tip

Electric currents in everyday circuits tend to be quite small, so it's common for examiners to throw in a **unit prefix** like 'm' next to quantities of current, e.g. 10 mA (10 milliamperes). Make sure you can convert these into standard units, e.g. $10 \text{ mA} = 10 \times 10^{-3} \text{ A}$.

Make sure to only use the triangle to help you **rearrange** the equation that links charge, current and time. Don't draw it if you are asked to write out the equation in full, such as $Q = I \times t$, as you may lose marks for doing so.

Check out this revision note on speed, distance and time if you need a reminder on how to use formula triangles.



Your notes



Your notes

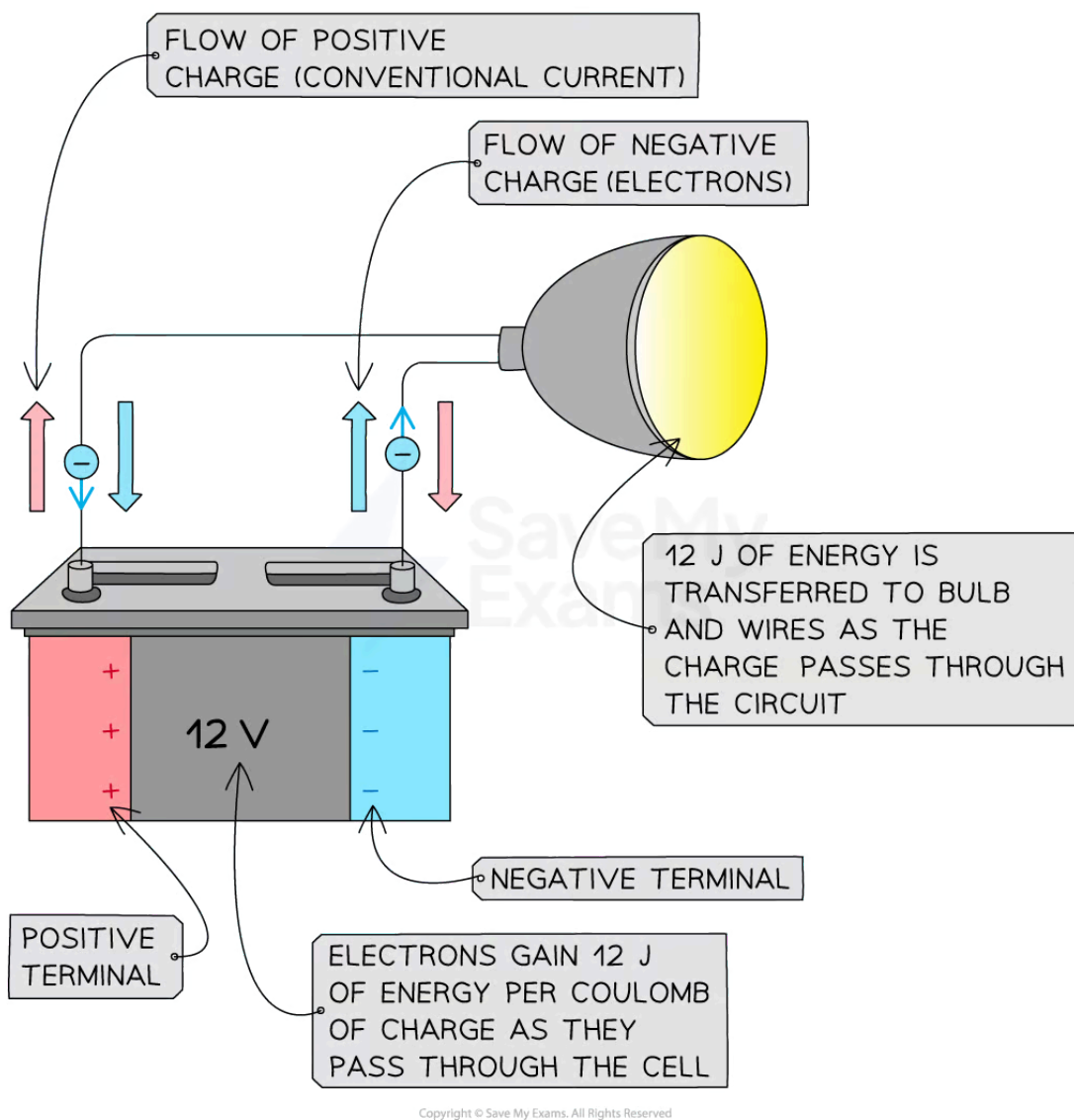
Voltage & Energy

Voltage

- Voltage is defined as
The energy transferred per unit charge passing between two points
- Voltage is measured in units of **volts (V)**
 - 1 volt is equivalent to the transfer of 1 joule of energy by 1 coulomb of charge, or $1\text{ V} = 1\text{ J/C}$
- The terminals of a cell make one end of the circuit **positive** and the other **negative**
- As electrons flow through a cell, they **gain** energy
 - For example, in a 12 V cell, every coulomb of charge passing through gains 12 J of energy
- As electrons flow through a circuit, they **lose** energy
 - For example, after leaving the 12 V cell, each coulomb of charge will transfer 12 J of energy to the wires and components in the circuit



Your notes



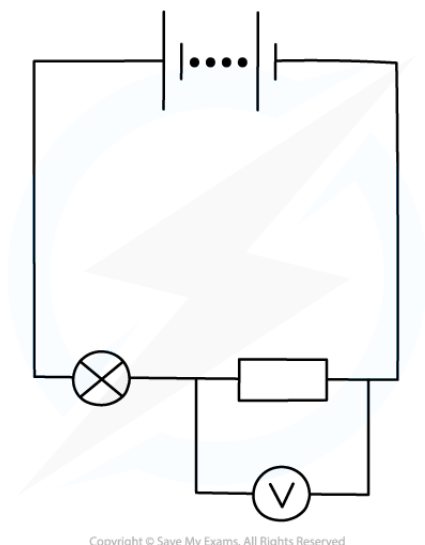
Electrons gain energy as they pass through a cell. As they flow through the light bulb, energy is transferred to the surroundings by heating and radiation

Measuring voltage

- Voltage can be measured using a **voltmeter**
- Voltmeters must be set up in **parallel** with the component being measured



Your notes



Voltage can be measured by connecting a voltmeter in parallel between two points in a circuit

Examiner Tip

When you are building a circuit in class, always **connect the voltmeter last**. Make the whole circuit first and check it works, and then connect the voltmeter so that the leads are on each side of the component you are measuring. This will save you a lot of time waiting for your teacher to troubleshoot your circuit!

You might sometimes see voltage called potential difference. This term can be useful when thinking about voltmeters as the potential difference describes a **difference** between **two** points, therefore the voltmeter has to be connected between **two** points in the circuit.



Your notes

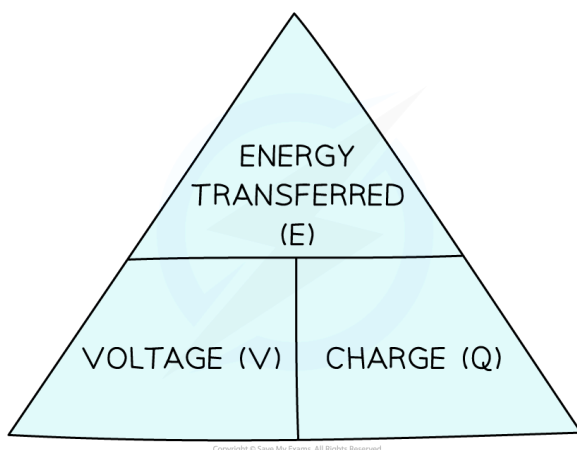
Calculating Voltage

- The equation linking the energy transferred, voltage and charge is given below:
energy transferred = charge $\times$ voltage

$$E = Q \times V$$

- Where:
 - E = energy transferred, measured in joules (J)
 - Q = charge moved, measured in coulombs (C)
 - V = voltage, measured in volts (V)
- This can be rearranged using the formula triangle below:

Energy charge voltage formula triangle



Formula triangle for the energy transferred, voltage and charge equation

- Check out this revision note on [speed, distance and time](#) if you need a reminder on how to use formula triangles



Your notes

 Worked example

The normal operating voltage for a lamp is 6 V.

Calculate how much energy is transferred in the lamp when 4200 C of charge flows through it.

Answer:

Step 1: List the known quantities

- Voltage, $V = 6 \text{ V}$
- Charge, $Q = 4200 \text{ C}$

Step 2: State the equation linking potential difference, energy and charge

- The equation linking potential difference, energy and charge is:

$$E = Q \times V$$

Step 3: Substitute the known values and calculate the energy transferred

$$E = 6 \times 4200 = 25\,200 \text{ J}$$

- Therefore, **25 200 J** of energy is transferred in the lamp

 Examiner Tip

Don't be confused by the symbol for voltage (the **symbol** V) being the same as its unit (the **volt**, V). Learn the equation and remember especially that one volt is equivalent to 'a joule per coulomb'.



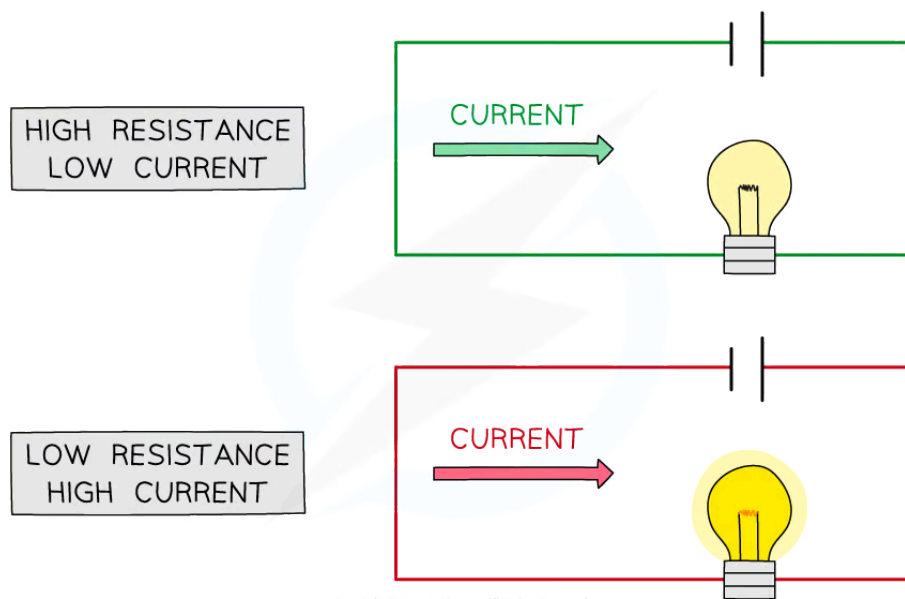
Your notes

Resistance

Calculating current, resistance & potential difference

- Resistance is defined as
The opposition of a component to the flow of electric current through it
- Resistance is measured in units of **ohms (Ω)**
 - A resistance of $1\ \Omega$ is equivalent to a voltage across a component of 1 V which produces a current of 1 A through it
- The resistance of a component controls the size of the current in a circuit
- For a given voltage across a component:
 - The **higher** the resistance, the **lower** the current that can flow
 - The **lower** the resistance, the **higher** the current that can flow
- All electrical components, including wires, have some value of resistance
- Wires are often made from copper because it has a **low** electrical resistance
 - This is why it is known as a **good conductor**

Comparing current and resistance



Copyright © Save My Exams. All Rights Reserved

A greater resistance means there is a lower current and vice versa

- The current, resistance and potential difference of a component in a circuit are calculated using the equation:

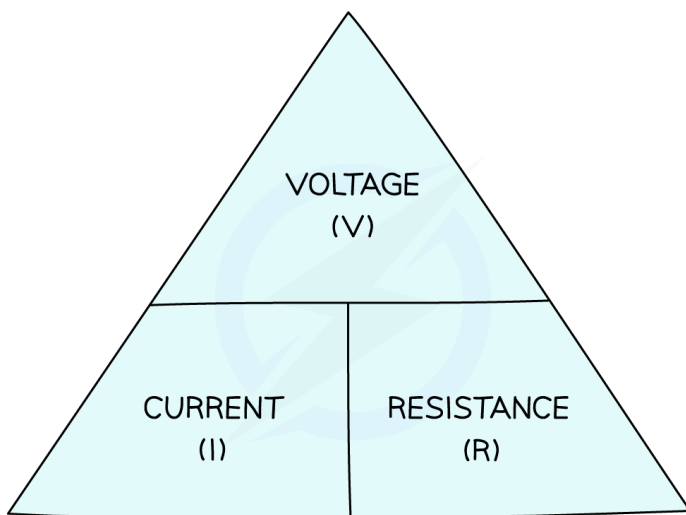
$$\text{voltage} = \text{current} \times \text{resistance}$$

$$V = I \times R$$



Your notes

- Where:
 - V = voltage, measured in volts (V)
 - I = current, measured in amps (A)
 - R = resistance, measured in ohms (Ω)
- This equation can be rearranged with the help of the following formula triangle:

Voltage current resistance formula triangle

Copyright © Save My Exams. All Rights Reserved

Formula triangle for the voltage, current and resistance equation

- Check out this revision note on [speed, distance and time](#) if you need a reminder on how to use formula triangles

**Worked example**

Calculate the voltage across a resistor of resistance $10\ \Omega$ if there is a current of $0.3\ \text{A}$ through it.

Answer:

Step 1: List the known quantities

- Resistance, $R = 10\ \Omega$
- Current, $I = 0.3\ \text{A}$

Step 2: Write the equation relating resistance, potential difference and current

$$V = I \times R$$

Step 3: Substitute in the values

$$V = 0.3 \times 10 = 3\ \text{V}$$

 Examiner Tip

In exam questions, the resistance of the wires, batteries, ammeters and voltmeters are always assumed to be **zero** (in the case of voltmeters, they have extremely high resistances so that current does not flow through them, and this has a negligible effect on the overall resistance of the circuit)



Your notes